

Question 1:	Marks
(a) Find (i) $\int \frac{x-2}{x^2+1} dx$	2
(ii) $\int \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx$	1
(b) Find $\int \frac{dx}{\sqrt{6-x-x^2}}$	3
(c) Evaluate $\int_0^{\frac{\pi}{4}} \tan^3 \theta d\theta$	3
(d) If $I_n = \int x^n \sin x dx$, show that	
(i) $I_n + n(n-1)I_{n-2} = x^{n-1}(n \sin x - x \cos x)$	3
(ii) Hence, evaluate $\int_0^{\pi} x^2 \sin x dx$	3
Question 2:	
(START A NEW PAGE)	
(a) Using the separate graphs of $y = f(x)$ provided at the end of the examination paper, sketch the graphs of the following on separate diagrams. Clearly label the coordinates of any intercepts with the coordinate axes and the position of any asymptotes.	
(i) $y = \frac{1}{f(x)}$	2
(ii) $y = f(2-x)$	3
(iii) $y^2 = f(x)$	2
(b) Given that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$, show that $\int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x + \cos x} dx = \frac{\pi}{4}$	3
(c) Find the centre of the circle that passes through the foci of the conics $4x^2 + 9y^2 = 1$ and $y = 2x^2$.	5

Question 3:	(START A NEW PAGE)	Marks
(a)	A function $y = f(x)$ has the following properties:	5
(b)	$f(0)=13$, $f(x) \rightarrow 0$ as $x \rightarrow -\infty$ and $f'(x) = e^x(x^2 - 5x + 6)$	
Draw a sketch of a function with these properties, clearly showing the positions of any stationary points, inflexion points and asymptotes.		
(c)	If $x^2 + 4y^2 = 24$ defines the equation of an ellipse	4
(i)	Find the coordinates of the foci and intercepts with the coordinate axes.	
(ii)	Find the equations of the directrices.	1
(iii)	Draw a neat sketch of the ellipse showing the above features.	3
(d)	Derive the equation of the tangent to the hyperbola $\frac{x^2}{2} - \frac{y^2}{3} = -4$ at the point $P(4,6)$.	2

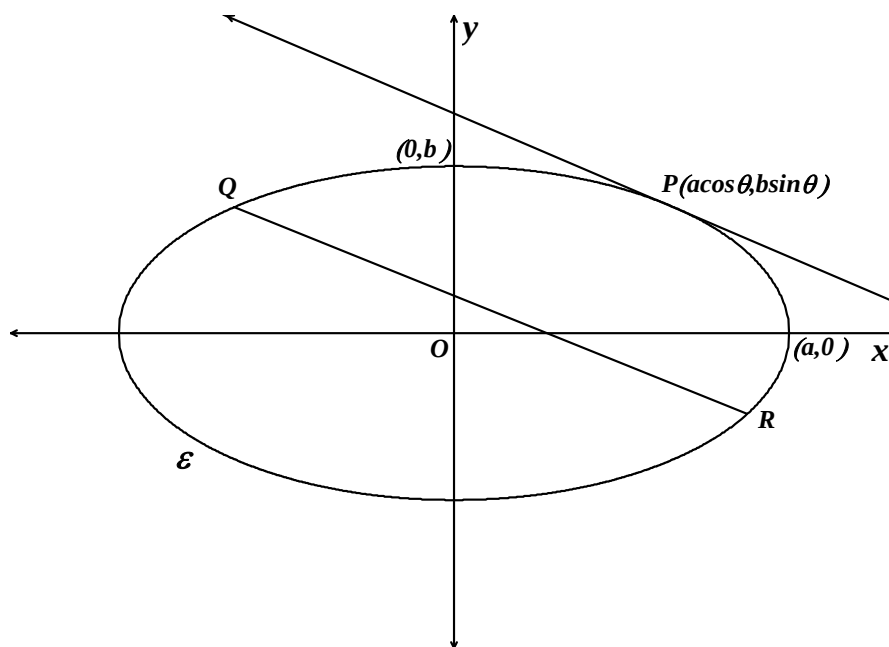
Question 4: (START A NEW PAGE)

(a)	The line $y = 2x - 4$ meets the hyperbola $\frac{x^2}{3} - \frac{y^2}{2} = 1$ at points M and N .	2
(i)	Find the size of the acute angle between the asymptotes of the given hyperbola.	
(ii)	Find the coordinates of the point of intersection of the tangent drawn from M and N .	3
(b)	(i) Find values for A , B and C so that $\frac{32}{16-x^4} \equiv \frac{A}{4+x^2} + \frac{B}{2+x} + \frac{C}{2-x}$	3
(ii)	Hence, evaluate $\int_0^1 \frac{dx}{16-x^4}$	3

QUESTION 4(c) is continued on the next page

QUESTION 4 (continued)

- (c) Consider the ellipse ε with equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, and the points $P(a \cos \theta, b \sin \theta)$, $Q(a \cos(\theta + \phi), b \sin(\theta + \phi))$ and $R(a \cos(\theta - \phi), b \sin(\theta - \phi))$ on ε .



- (i) Prove that the equation of the tangent to ε at the point P is $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$. 2
- (ii) Show that the chord QR is parallel to the tangent at P . 2

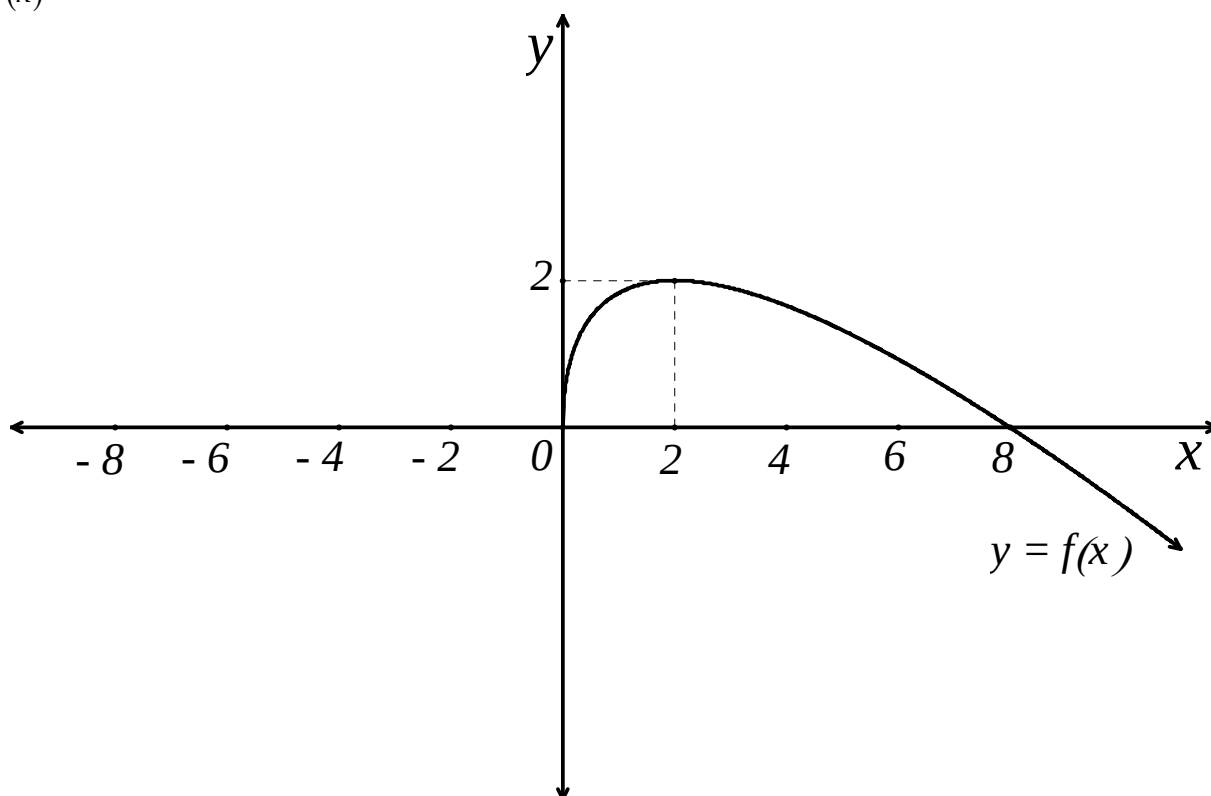
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QUESTION 2(a)(i), (ii)

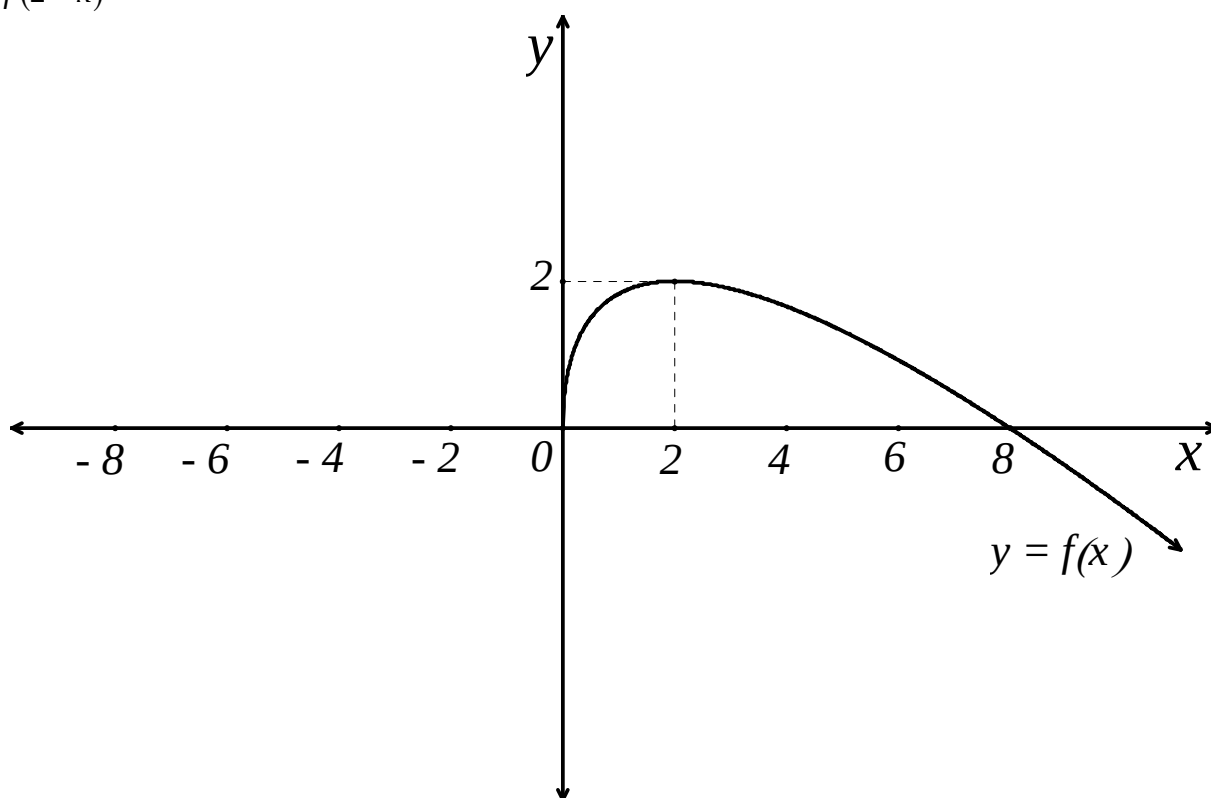
STUDENT NUMBER:

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(i) $y = \frac{1}{f(x)}$



(ii) $y = f(2 - x)$

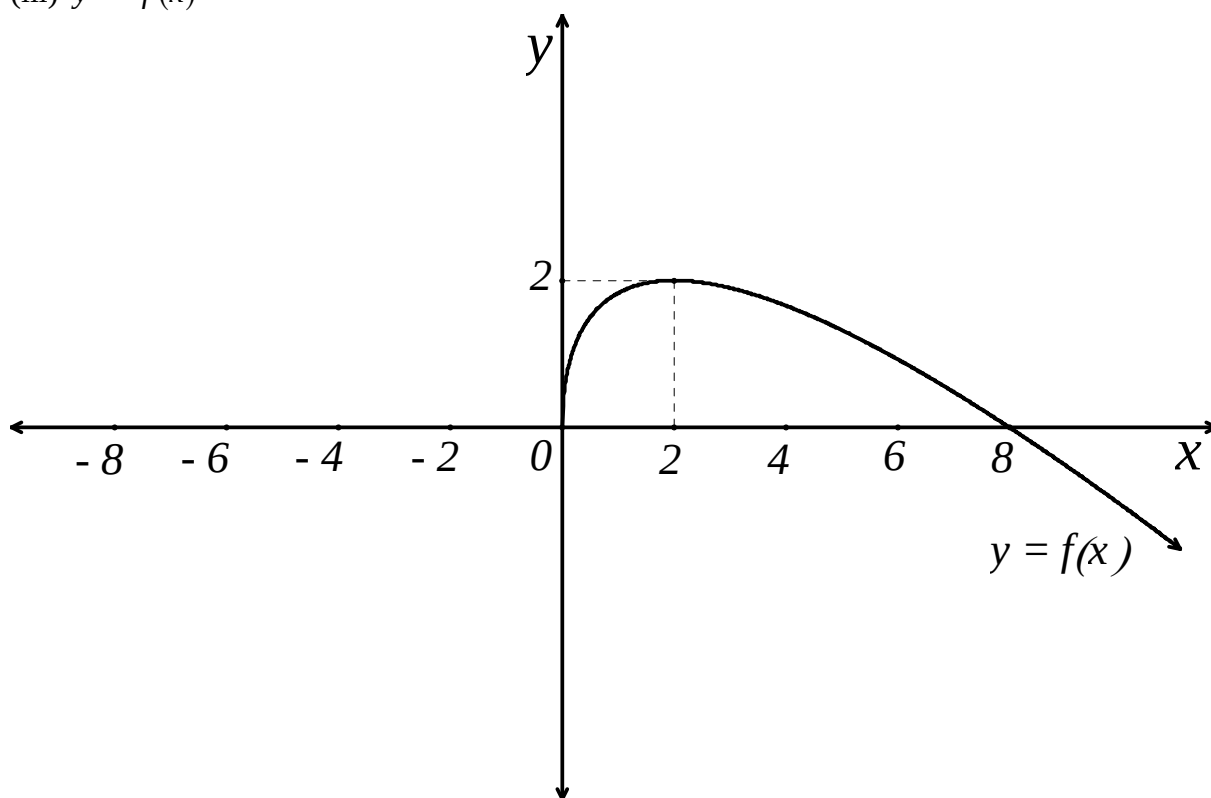


QUESTION 2(a)(iii)

STUDENT NUMBER:

INCLUDE THESE SHEETS WITH YOUR ANSWERS TO Q2

(iii) $y^2 = f(x)$

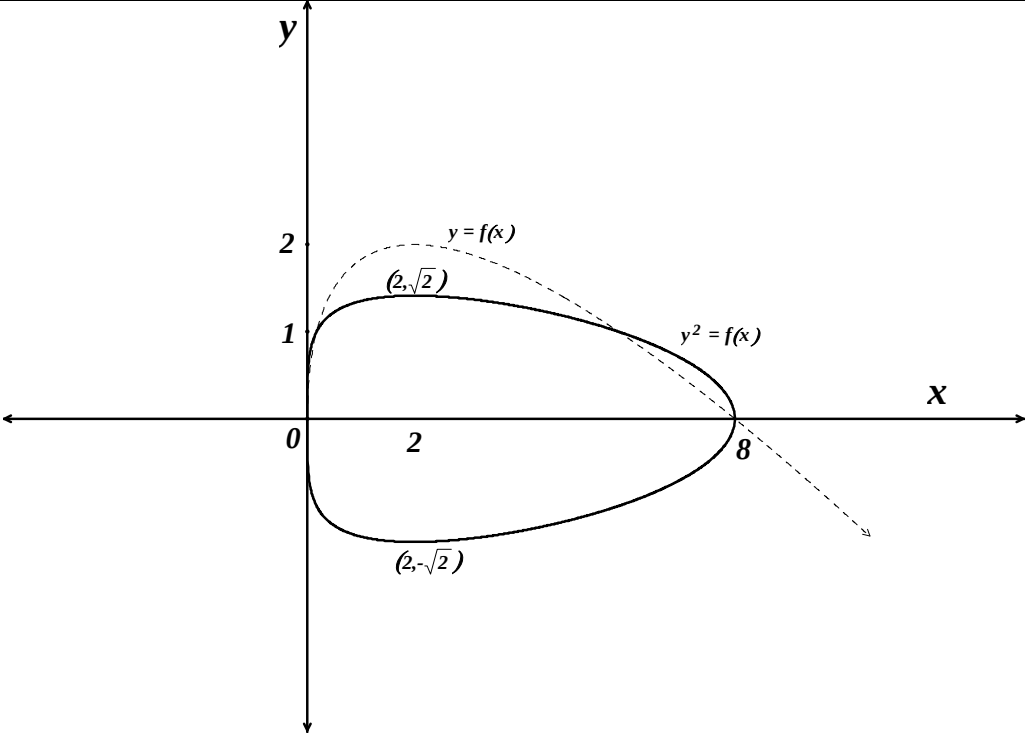


QUESTION 1			MARKS
(a)	(i)	$\int \frac{x-2}{x^2+1} dx = \int \left(\frac{x}{x^2+1} - \frac{2}{x^2+1} \right) dx$ $= \frac{1}{2} \ln(x^2+1) - 2 \tan^{-1} x + c$	
	(ii)	$\int \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx = \frac{1}{2} (\sin^{-1} x)^2 + c$	
(b)		$\int \frac{dx}{\sqrt{6-x-x^2}} = \int \frac{dx}{\sqrt{\left(\frac{5}{2}\right)^2 - \left(x+\frac{1}{2}\right)^2}}$ $= \sin^{-1} \left(\frac{x+\frac{1}{2}}{\frac{5}{2}} \right) + c$ $= \sin^{-1} \left(\frac{2x+1}{5} \right) + c$	1 1 1
(c)		$\int_0^{\frac{\pi}{4}} \tan^3 \theta d\theta = \int_0^{\frac{\pi}{4}} \tan \theta (\sec^2 \theta - 1) d\theta$ $= \int_0^{\frac{\pi}{4}} (\sec^2 \theta \tan \theta - \tan \theta) d\theta$ $= \left[\frac{1}{2} \tan^2 \theta + \ln(\cos \theta) \right]_0^{\frac{\pi}{4}}$ $= \left(\frac{1}{2} \tan^2 \left(\frac{\pi}{4} \right) + \ln \left(\cos \left(\frac{\pi}{4} \right) \right) \right) - \left(\frac{1}{2} \tan^2(0) + \ln(\cos(0)) \right)$ $= \frac{1}{2} + \ln \left(\frac{1}{\sqrt{2}} \right) \quad \left[\text{or} \quad = \frac{1}{2} (1 - \ln 2) \right]$ Or	

	$ \begin{aligned} \int_0^{\frac{\pi}{4}} \tan^3 \theta \, d\theta &= \int_0^{\frac{\pi}{4}} \tan \theta (\sec^2 \theta - 1) \, d\theta \\ &= \int_0^{\frac{\pi}{4}} (\sec^2 \theta \tan \theta - \tan \theta) \, d\theta \\ &= \int_0^{\frac{\pi}{4}} \{(\sec \theta \tan \theta) \sec \theta - \tan \theta\} \, d\theta \\ &= \left[\frac{1}{2} \sec^2 \theta + \ln(\cos \theta) \right]_0^{\frac{\pi}{4}} \\ &= \left(\frac{1}{2} \sec^2 \left(\frac{\pi}{4} \right) + \ln \left(\cos \left(\frac{\pi}{4} \right) \right) \right) - \left(\frac{1}{2} \sec^2(0) + \ln(\cos(0)) \right) \\ &= 1 + \ln \left(\frac{1}{\sqrt{2}} \right) - \frac{1}{2} \\ &= \frac{1}{2} - \frac{1}{2} \ln 2 \quad \left[\text{or} \quad = \frac{1}{2} (1 - \ln 2) \right] \end{aligned} $	
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(d)	(i)	$I_n = \int x^n \sin x \, dx$ $= \int x^n \cdot \frac{d}{dx}(-\cos x) \, dx$ $= [x^n(-\cos x)] - \int (-\cos x) \cdot nx^{n-1} \, dx$ $= -x^n \cos x + n \int x^{n-1} \frac{d}{dx}(\sin x) \, dx$ $= -x^n \cos x + n \left[x^{n-1} \sin x - \int (n-1)x^{n-2} \sin x \, dx \right]$ $= -x^n \cos x + nx^{n-1} \sin x - n(n-1) \int x^{n-2} \sin x \, dx$ $= -x^n \cos x + nx^{n-1} \sin x - n(n-1)I_{n-2}$ $\therefore I_n + n(n-1)I_{n-2} = x^{n-1}(n \sin x - x \cos x)$	1 1 1
	(ii)	$\int_0^\pi x^2 \sin x \, dx = I_2$ $I_n = [x^{n-1}(n \sin x - x \cos x)]_0^\pi - n(n-1)I_{n-2}$ $I_2 = [x(2 \sin x - x \cos x)]_0^\pi - 2I_0$ $I_0 = \int_0^\pi \sin x \, dx$ $= [-\cos x]_0^\pi$ $= -\cos \pi + \cos 0$ $= 2$ $I_2 = [x(2 \sin x - x \cos x)]_0^\pi - 2I_0$ $= \{\pi(2 \sin \pi - \pi \cos \pi) - 0\} - 2\{2\}$ $= \pi^2 - 4$	

QUESTION 2			MARKS
(a)	(i)	Graph showing the function $y = f(x)$ (solid curve) and its reciprocal $y = \frac{1}{f(x)}$ (dashed curve). The function $y = f(x)$ has a vertical asymptote at $x = 0$ and a horizontal asymptote at $y = 0$. The reciprocal function $y = \frac{1}{f(x)}$ has a vertical asymptote at $x = 8$ and a horizontal asymptote at $y = 0$. The point $(2, 0.5)$ is marked on the curve $y = f(x)$.	
	(ii)	Graph showing the function $y = f(x)$ (dashed curve) and its reflection $y = f(2-x)$ (solid curve). The function $y = f(x)$ has a vertical asymptote at $x = 0$ and a horizontal asymptote at $y = 0$. The reflection $y = f(2-x)$ has a vertical asymptote at $x = 2$ and a horizontal asymptote at $y = 0$. The graph is the reflection of $y = f(x)$ in the line $x = 1$. The point $(-6, 0)$ is marked on the curve $y = f(2-x)$.	

Q2 (a)	(iii)		
(b)	$\int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x + \cos x} dx = \int_0^{\frac{\pi}{2}} \frac{\sin(\frac{\pi}{2} - x)}{\sin(\frac{\pi}{2} - x) + \cos(\frac{\pi}{2} - x)} dx$ $= \int_0^{\frac{\pi}{2}} \frac{\cos x}{\cos x + \sin x} dx$ Let $I = \int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x + \cos x} dx$ $\therefore 2I = \int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x + \cos x} dx + \int_0^{\frac{\pi}{2}} \frac{\cos x}{\cos x + \sin x} dx$ $= \int_0^{\frac{\pi}{2}} \frac{\sin x + \cos x}{\sin x + \cos x} dx$ $= \int_0^{\frac{\pi}{2}} 1 dx$ $= \frac{\pi}{2}$ $I = \frac{\pi}{4}$		
(c)	$\frac{x^2}{\frac{1}{4}} + \frac{y^2}{\frac{1}{9}} = 1 \quad \text{where } a = \frac{1}{2}, \quad b = \frac{1}{3}$ $b^2 = a^2(1 - e^2)$ $\frac{1}{9} = \frac{1}{4}(1 - e^2)$ $4 = 9(1 - e^2)$ $4 = 9 - 9e^2$ $9e^2 = 5$ $e^2 = \frac{5}{9}$		

$$e = \frac{\sqrt{5}}{3}$$

Foci $(\pm ae, 0) = (\pm \frac{1}{2} \cdot \frac{\sqrt{5}}{3}, 0)$
 $= (\pm \frac{\sqrt{5}}{6}, 0)$

Parabola $x^2 = \frac{1}{2}y$
 $= 4(\frac{1}{8})y$

$\therefore$ focus $(0, \frac{1}{8})$

Let coordinates of centre be: $C(0, -k)$
 Now $SC = BC$ (both radii)

$$\frac{1}{8} + k = \sqrt{\left(\frac{\sqrt{5}}{6}\right)^2 + k^2}$$

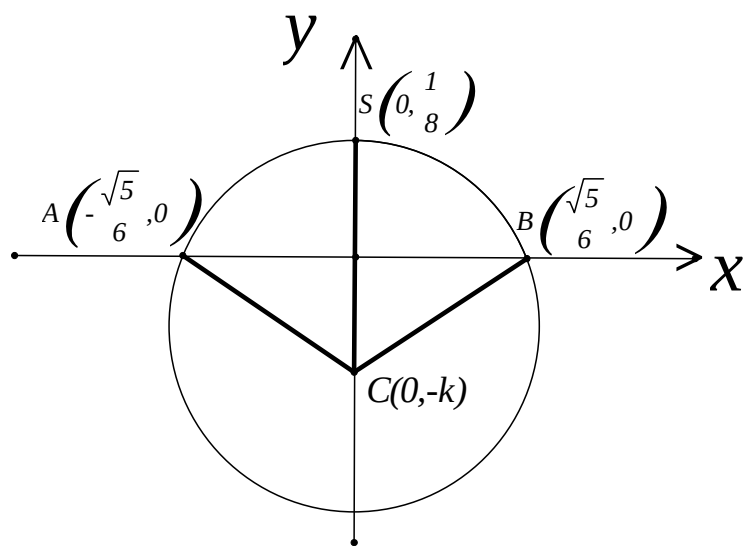
$$\left(\frac{1}{8} + k\right)^2 = \frac{5}{36} + k^2$$

$$\frac{1}{64} + \frac{1}{4}k + k^2 = \frac{5}{36} + k^2$$

$$\frac{1}{4}k = \frac{71}{576}$$

$$k = \frac{71}{144}$$

$\therefore$ coordinates of centre is $\left(0, -\frac{71}{144}\right)$



QUESTION 3

- (a) at $x = 0$, $f'(x) = 6$
 as $x \rightarrow \infty$ $f'(x) \rightarrow \infty$ $\therefore$ gets steeper
 as $x \rightarrow -\infty$ $f'(x) \rightarrow 0$ $\therefore$ graph flattens out
 for $x < 2$ or $x > 3$, $x^2 - 5x + 6 > 0$, $\therefore f'(x) > 0$
 for $2 < x < 3$, $x^2 - 5x + 6 < 0$, $\therefore f'(x) < 0$
 for $x = 2$ or 3 , $f'(x) = 0$, $\therefore$ stat. pts.

$$f''(x) = e^x(x^2 - 3x + 1)$$

For possible inflexion points $f''(x) = 0$

$$e^x(x^2 - 3x + 1) = 0$$

$$x = \frac{3 \pm \sqrt{5}}{2} \quad (e^x \neq 0)$$

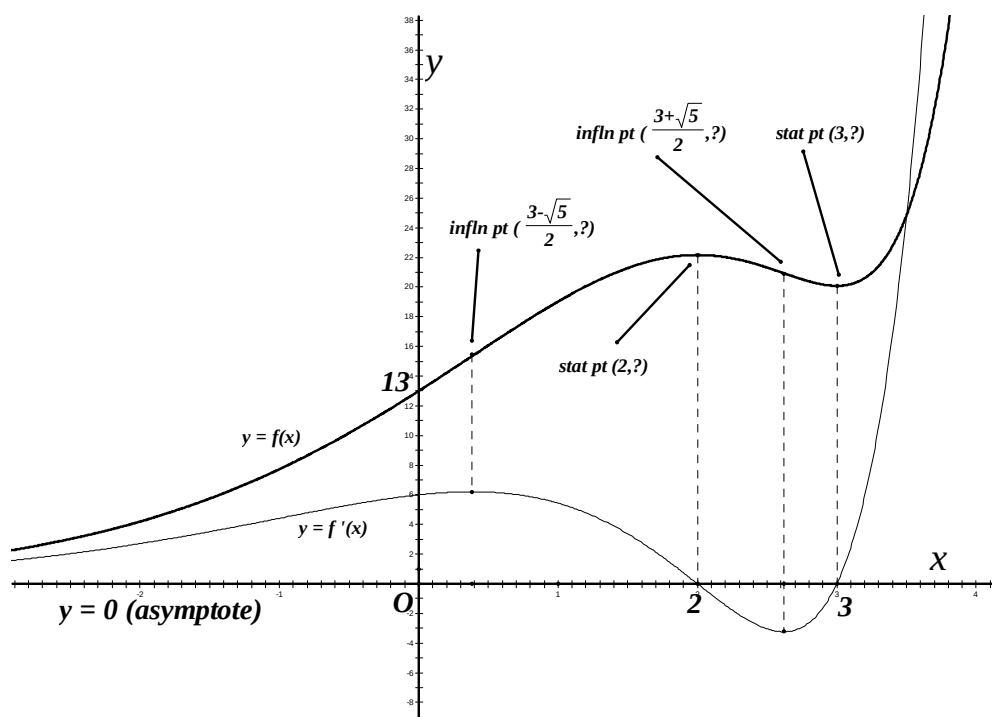
Test concavity change

x	0.3	$\frac{3 - \sqrt{5}}{2} \approx 0.38$	0.4
$f''(x)$	≈ 0.26 > 0	0	≈ -0.06 < 0

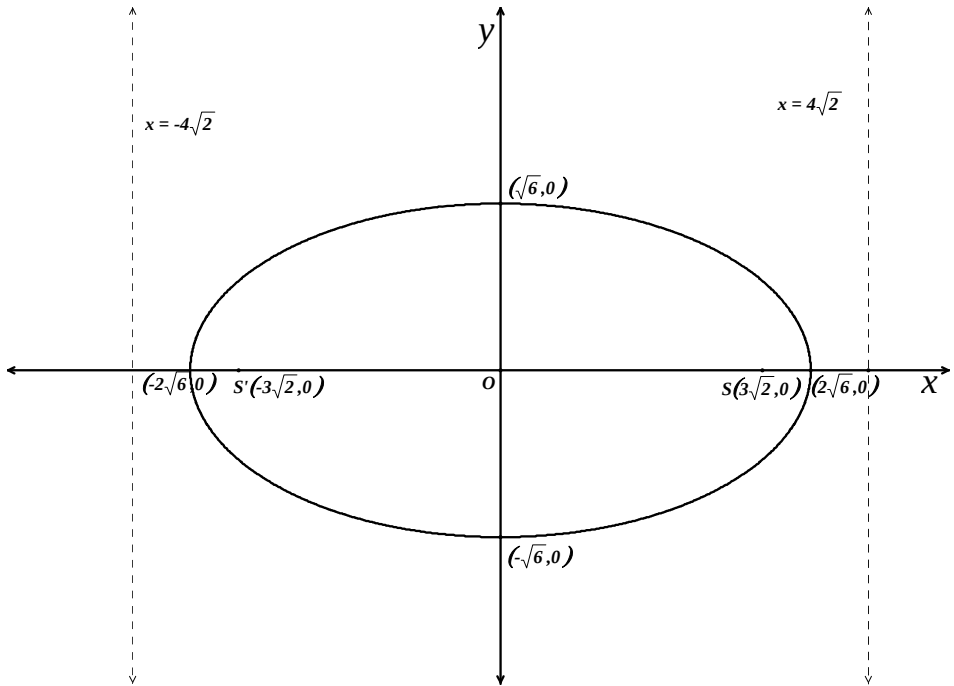
change in concavity, $\therefore$ inflexion point

x	2.5	$\frac{3 + \sqrt{5}}{2} \approx 2.62$	2.7
$f''(x)$	≈ -3.05 < 0	0	≈ 2.83 > 0

change in concavity, $\therefore$ inflexion point



(b)	(i)	$\frac{x^2}{24} + \frac{y^2}{6} = 1 \quad \Rightarrow \quad a = 2\sqrt{6} \quad \text{and} \quad b = \sqrt{6}$ From $b^2 = a^2(1 - e^2)$, we get $e = \sqrt{1 - \frac{b^2}{a^2}}$ and hence $e = \sqrt{1 - \frac{6}{24}} = \frac{\sqrt{3}}{2}$ ▀ Foci given by $(\pm ae, 0) = (\pm 3\sqrt{2}, 0)$ and coordinates of vertices/intercepts with axes $(\pm 2\sqrt{6}, 0)$ and $(0, \pm\sqrt{6})$	
	(ii)	$x = \pm \frac{a}{e}$ $= \pm \frac{2\sqrt{6}}{\frac{1}{2}\sqrt{3}}$ $= \pm 4\sqrt{2}$	

Q3 (b)	(iii)			
(c)	gradient: $\frac{2x}{2} - \frac{2y}{3} \frac{dy}{dx} = 0$ $\therefore \frac{dy}{dx} = \frac{3x}{2y}$ at (4,6) $\frac{dy}{dx} = \frac{3 \times 4}{2 \times 6}$ = 1 Equation of tangent: $y - 6 = 1(x - 4)$ i.e. $x - y + 2 = 0$ is the required tangent.			

QUESTION 4		
(a)	(i)	$\frac{x^2}{3} + \frac{y^2}{2} = 1 \Rightarrow a = \sqrt{3} \text{ and } b = \sqrt{2}$ Equation of asymptotes: $y = \pm \frac{b}{a} x$ $y = \pm \frac{\sqrt{2}}{\sqrt{3}} x$ Hence slopes of these asymptotes are $m_1 = \frac{\sqrt{2}}{\sqrt{3}}$ and $m_2 = -\frac{\sqrt{2}}{\sqrt{3}}$ Let angle between asymptote $y = \frac{\sqrt{2}}{\sqrt{3}} x$ and x-axis be θ then $\tan \theta = \sqrt{\frac{2}{3}}$ and required angle = 2θ $\text{angle} = 2 \tan^{-1} \left(\sqrt{\frac{2}{3}} \right)$ $= 78^\circ \text{ (to nearest degree)}$
	(ii)	Let $P(x_0, y_0)$ be an external point to the point of contact of tangents at M and N. Then the chord of contact is of the form: $\frac{x_0 x}{3} - \frac{y_0 y}{2} = 1$ i.e. $2x_0 x - 3y_0 y = 6$ Comparing $2x - y = 4$ and $2x_0 x - 3y_0 y = 6$ $\frac{2x_0}{2} = \frac{6}{4} \text{ and } \frac{3y_0}{1} = \frac{6}{4}$ $x_0 = 1\frac{1}{2} \text{ and } y_0 = \frac{1}{2}$ Point is $\left(1\frac{1}{2}, \frac{1}{2}\right)$

(b)	(i)	$\frac{32}{16-x^4} = \frac{A(2+x)(2-x) + B(4+x)(2-x) + C(4+x^2)(2+x)}{(4+x^2)(2-x)(2+x)}$ $32 \equiv A(2+x)(2-x) + B(4+x^2)(2-x) + C(4+x^2)(2+x)$ when $x = 2$, $32 = 32C$ $C = 1$ when $x = -2$, $32 = 32B$ $B = 1$ when $x = 0$, $32 = 4A + 8B + 8C$ $A = 4$ $\therefore \frac{32}{16-x^4} = \frac{4}{(4+x^2)} + \frac{1}{(2-x)} + \frac{1}{(2+x)}$	
	(ii)	$\int_0^1 \frac{dx}{16-x^4} = \frac{1}{32} \int_0^1 \left(\frac{4}{4+x^2} + \frac{1}{2-x} + \frac{1}{2+x} \right) dx$ $= \frac{1}{32} \left[2 \tan^{-1} \left(\frac{x}{2} \right) - \ln(2-x) + \ln(2+x) \right]_0^1$ $= \frac{1}{32} \left\{ \left(2 \tan^{-1} \left(\frac{1}{2} \right) - \ln(1) + \ln(3) \right) - \left(2 \tan^{-1}(0) - \ln(2) + \ln(2) \right) \right\}$ $= \frac{1}{32} \left(2 \tan^{-1} \left(\frac{1}{2} \right) + \ln(3) \right)$	
(c)	(i)	$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$ $\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$ at $P(a \cos \theta, b \sin \theta)$, $\frac{dy}{dx} = -\frac{b^2(a \cos \theta)}{a^2(b \sin \theta)}$ $= -\frac{b \cos \theta}{a \sin \theta}$ Tangent is $y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$ $(a \sin \theta)y - ab \sin^2 \theta = -(b \cos \theta)x + ab \cos^2 \theta$ $(b \cos \theta)x + (a \sin \theta)y = ab(\sin^2 \theta + \cos^2 \theta)$ $(b \cos \theta)x + (a \sin \theta)y = ab$ $\frac{(b \cos \theta)}{ab}x + \frac{(a \sin \theta)}{ab}y = \frac{ab}{ab}$ $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$	
	(ii)	slope of tangent $= -\frac{b \cos \theta}{a \sin \theta}$ $= -\frac{b}{a} \cot \theta$	

		$\begin{aligned} \text{slope of chord } QR &= \frac{b \sin(\theta + \phi) - b \sin(\theta - \phi)}{a \cos(\theta + \phi) - a \cos(\theta - \phi)} \\ &= \frac{b (\sin \theta \cos \phi + \cos \theta \sin \phi) - (\sin \theta \cos \phi - \cos \theta \sin \phi)}{a (\cos \theta \cos \phi - \sin \theta \sin \phi) - (\cos \theta \cos \phi + \sin \theta \sin \phi)} \\ &= \frac{b \left(\frac{2 \cos \theta \sin \phi}{-2 \sin \theta \sin \phi} \right)}{a} \\ &= -\frac{b}{a} \cot \theta \\ \therefore \text{ tangent is parallel to chord (equal slopes)} \end{aligned}$	
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